

MTH302

MIDTERM BY VU SPARKY



Version (1.1)

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1. A book sells for Rs. 50. It has just been reduced 40%. What will be the selling price?

- a. Rs. 40
- b. **Rs. 30** ✓^{HB}
- c. Rs. 10
- d. Rs. 20

2. Which of the following statement is the best option for a column matrix?

- a. It has only one row.
- b. It has no column.
- c. **It has only one column.** ✓^{HB}
- d. It has no row.

3. MINVERSE returns the ----- for the matrix stored in an array.

- a. Scalar matrix
- b. **Inverse matrix** ✓^{HB}

4. The amount added to a cost price while calculating a selling price is called

- a. margin
- b. **markup** ✓^{HB}
- c. markdown
- d. actual price

5. Atif and Bilal won first and second places in a music contest. They have to share the cash prize in 7:2. If Atif received 35 \$. How much more money did Atif received than Bilal.

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a. **25**✓^{HB}

b. 35

c. 5

d. 10

6. The ratio of the weight of Tina's book to the weight of Jean's book is 5:6 .Tina's book weighs 30 ounces. How much less does Tina's book weigh than Jean's book?

a. 72

b. 12

c. **6**✓^{HB}

d. 36

7. If A, B and C are matrices with orders 3×3 , 2×3 and 4×2 respectively. Then which of the following is the order of CBA?

a. 3×3

b. 4×2

c. **4×3** ✓^{HB}

d. 2×3

8. If $42 : 6 = x : 5$, what is the value of x?

a. 32

b. 38

c. **35**✓^{HB}

d. 30

9. The correct relation among %markup on sale, cost price and selling price is...

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- a. **Selling Price = Cost price + (cost price × %Markup on sale)** ✓^{HB}
- b. Selling Price = Cost price + (Selling price × %Markup on sale)
- c. None of these
- d. Cost Price = Cost price + (Selling price × %Markup on sale)
10. Ahmed got 65 marks in mathematics in the midterm exam. If he gets 80 marks in the final term exam in the same subject, what is the percentage increase in the marks?
- a. **23.1%** ✓^{HB}
- b. 26.2%
- c. 20.5%
- d. 25.5%
11. The value of unknown x in the proportion $2x : 2 = (x+1) : 5$ is
- a. $5/2$
- b. **$1/4$** ✓^{HB} @studywithhamza25
- c. $1/2$
- d. 4
12. Which of the following statement is the best option for a row matrix?
- a. It has only one column.
- b. **It has only one row.** ✓^{HB}
- c. It has no column.
- d. It has no row.

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13. If SC is the current selling price and SN is the new selling price and M is the markdown then which is the correct formula to find the %age markdown

- a. $(SC - SN) / SN * 100$
- b. $(SC - SN) / SC * 100$ ✓^{HB}
- c. $(SC - SN) / SN$
- d. $(SC - SN) / SC$

14. Which of the following is true for the following matrix.

15. [257]

- a. It is a square matrix
- b. It is a row matrix ✓^{HB}
- c. It is a column matrix
- d. It is an identity matrix

16. Which of the following statement about the addition of two matrices is true?

- a. Number of rows of first matrix is equal to the number of rows of the second matrix
- b. Number of columns and rows of first matrix is equal to the number of columns and rows of the second matrix ✓^{HB}
- c. Number of columns of first matrix is equal to the number of columns of the second matrix

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- d. Number of rows of first matrix is equal to the number of columns of the second matrix
17. If the current selling price is 2200 and %Markdown is 20 then what is the new selling price
- a. $=2200/(0.2*2200)$
 - b. $=2200-(0.2*2200)$ ✓^{HB}
 - c. $=(0.2*2200)/2200$
 - d. $=2200+(0.2*2200)$
18. What is the Cost if selling price is Rs. 100 and its markup is Rs. 23?
- a. 77 ✓^{HB}
 - b. 123
 - c. 59
 - d. None of these
19. To find the inverse of a matrix A in excel, which one of the following formula can we use?
- a. $=\text{MINVERSE}(C4:D5)$ ✓^{HB}
 - b. $=\text{MINVERSE}(C4,D5)$
 - c. $=\text{MINVERS}(C4:D5)$
 - d. $=\text{MINVERSE}(C4-D5)$
20. What is the value of x in the proportion $3:2x = 5:2$?
- a. $2/5$
 - b. $3/5$ ✓^{HB}

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c. $1/5$

d. $4/5$

21. If matrix Y is a 4×4 matrix and matrix Z is a 4×3 matrix. Then, why the product AB exists?

a. Because of the No. of rows of Y is not equal to the No. of columns of Z.

b. Because of the No. of columns of Y is equal to the No. of rows of Z. ✓^{HB}

c. Because of the No. of columns of Y is not equal to the No. of columns of Z.

d. Because of the No. of rows of Y is equal to the No. of rows of Z.

22. In the word BALLOONS, the ratio of vowels to consonants is

a. $3/8$

b. $8/5$

c. $5/3$

d. $3/5$ ✓^{HB}

23. If $4:7 = x:5$, then what is x ?

a. $16/7$

b. $20/7$ ✓^{HB}

c. $19/7$

d. $15/7$

24. If $4/5$ students in a university are on vacations, how many out of every 100 students were on vacations?

a. 80 Students ✓^{HB}

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- b. 8 Students
- c. 1 Student
- d. 20 Students

25. More than one discount may be allowed on an item. In such case, first discount is allowed on the total price of the article and then the second discount is allowed on the amount payable

- a. **after first discount** ✓^{HB}
- b. before first discount
- c. None of these
- d. both are offered at the same time

26. If a matrix M with dimension $n \times n$, where ' $n \geq 1$ ', then which of the following is true for M?

- a. It is a rectangular matrix.
- b. It is a row matrix.
- c. **It is a square matrix.** ✓^{HB}
- d. It is a column matrix.

27. Rs.700 is divided between A and B in the ratio of 3:4, What is the A's share?

- a. 450
- b. 350
- c. **300** ✓^{HB}
- d. 400

28. Which fraction can be used to form a proportion with $\frac{7}{8}$?

- a. $\frac{14}{15}$
- b. $\frac{13}{16}$

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c. **21/24** ✓^{HB}

d. 35/25

29. If 'I' is an identity matrix of order 2×2 and A is any matrix of order 2×2 . Then, which of the following is equal to AI and IA?

a. I

b. **A** ✓^{HB}

c. E

d. O

30. The value of unknown x in the proportion $2x : 5 = (x+1) : 10$ is _____.

a. None of these

b. Not possible

c. 0

d. **1/3** ✓^{HB}

31. Which of the following is true for the calculation of the difference of two matrices?

a. Divide the corresponding elements of the matrices.

b. **Subtract the corresponding elements of the matrices.** ✓^{HB}

c. Multiply the corresponding elements of the matrices.

d. Add the corresponding elements of the matrices.

32. 12/45 is proportional to _____.

a. **2/15** ✓^{HB}

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- b. 3/15
c. None of these
d. 4/15
33. In order to stimulate the demand, the deduction in original sale price of an item is called
- a. markdown
b. **discount** ✓^{HB}
c. markup
d. retail price
34. Lucky bought a knife set through the school. She paid 500 Rs. after being given a 10% student discount. How much money did lucky save?
- a. **55 Rs.** ✓^{HB}
b. 454 Rs.
c. 555 Rs.
d. 450 Rs.
35. The relation between new selling price, %markdown and current selling price is:
- a. %markdown = %markdown*(1 - new selling price)
b. %markdown = %markdown*(1 - Current selling price)
c. **New selling price = Current selling price*(1 - %markdown)** ✓^{HB}
d. Current selling price = New selling price*(1 - %markdown)

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36. Under what conditions the inverse of 7 exists?

- a. $(7).(1/7) = (1/7).(7)=1$ ✓^{HB}
- b. $1/7$
- c. $(1/7).(7)$
- d. $(7).(1/7)$

37. Let A be a square matrix of dimension 3×3 (3 cross 3). What will be the dimension of matrix $2A$?

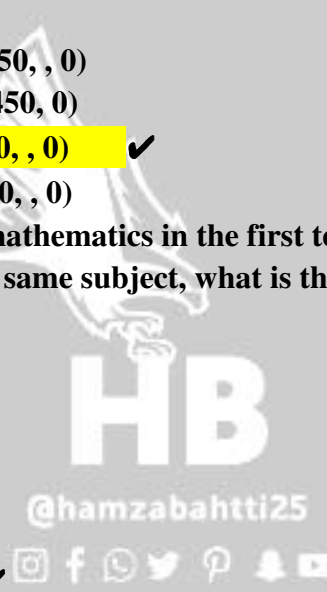
- a. 2×2
- b. 6×6
- c. 3×3 ✓^{HB}
- d. 2×3

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1. A Company signed a three year collective agreement that provided for wage increases of 5%, 3%, and 2% in successive years. An employee is currently earning 6500 rupees per month. What will be the salary per month at the end of the term of the contract?
 - a. Rs 7190.55
 - b. Rs 7180.45
 - c. Rs 7150.25
 - d. **Rs 7170.35 ✓**
2. The present value of an investment that pays Rs450 at the end of every month for 3 years. The money paid out will earn 9.5% annually. Which of the following is correct option?
 - a. $=PV(9.5\%/12, 3, 450, , 0)$
 - b. $=PV(9.5\%, 3*12, 450, 0)$
 - c. **$=PV(9.5\%/12, 3*12, 450, , 0)$ ✓**
 - d. $=FV(9.5\%/12, 3*12, 450, , 0)$
3. Ahmed got 65 marks in mathematics in the first term exam. If he gets 80 marks in the mid term exam in the same subject, what is the percentage increase in the marks?
 - a. **23.1% ✓**
 - b. 20.5%
 - c. 26.2%
 - d. 25.5%
4. An equation $2x+1 = 0$ is a 
 - a. **Linear equation ✓**
 - b. Cubic Equation
 - c. none of these
 - d. Quadratic equation
5. Net current asset value per share is
 - a. **$(\text{Current assets} - \text{Total liabilities}) / \text{No. of shares outstandings}$ ✓**
 - b. $(\text{Total liabilities} - \text{Current assets}) / \text{No. of shares outstandings}$
 - c. $(\text{Current assets} - \text{No. of shares outstandings}) / \text{Total liabilities}$
 - d. $(\text{No. of shares outstandings} - \text{Total liabilities}) / \text{Current assets}$
6. Carla wants to give her waiter a 15% tip. If her bill came to \$8.34, how much will she spend altogether?
 - a. 23.34
 - b. 8.94
 - c. 1.25

d. 9.59 ✓

7. Which of the following is NOT an arithmetic operator?

a. Multiplication (*)

b. Exponent (^)

c. Dollar (\$) ✓

d. Addition (+)

8. At a supermarket, a certain item has increased from 75 cents per pound to 81 cents per pound. What is the percentage change in the cost of the item?

a. $\{(81-75)/75\}\%$

b. $\{(81-75)/75\}*100\%$ ✓

c. $\{-(75-81)/81\}*100\%$

d. $\{-(75-81)/81\}\%$

9. Average = -----

a. number of data values/Sum of all data values

b. number of data values-Sum of all data values

c. Sum of all data values+number of data values

d. Sum of all data values/number of data values ✓

10. Gross remuneration is the amount after usual deductions such as taxes etc.

a. True

b. False ✓

11. Gross Salary is same as Take Home Salary.

a. True

b. False ✓

12. Aslam works in a supermarket for \$10.00 per hour. If his pay is increased to \$12.00, then what is his percentage increase in pay?

a. $\{(12-10)/12\}*100\%$

b. $\{(12+10)/12\}*100\%$

c. $\{(12+10)/10\}*100\%$

d. $\{(12-10)/10\}*100\%$ ✓

13. If you want to withdraw Rs 30,000 at the end of each 6 months for the next 5 years then what amount you must invest today at 7% compounded semi annually

a. $30,000 [((1 + .035)^{10}) + 1] / 0.035]$

b. $30,000 [(1 - (1 + .035)^{-10}) / .035]$ ✓

c. $30,000 [(1 - (1 + .07)^{-5}) / .07]$

d. $30,000 [((1 + .07)^5 - 1) / .07]$

14. In formula $PV = C*[1 - (1+i)^{-ni}] / i$, ni stands for

a. no of years

b. no of payments ✓

- c. no of months
d. none of these
15. It is asked to find a number whose 10% is 36. Which of the following is correct list of given data?
- a. Percentage = 10% ; Rate = 36
b. Percentage = 10% ; Base = 36
c. Rate = 10% ; Base = 36
d. Rate = 10% ; Percentage = 36 ✓
16. If you wish to have Rs. 15,0000 after 15 years from now, what will you need to invest this year if you earn 8% interest compounded quarterly?
- a. $15,0000 (1 + 0.08)^{15}$
b. $15,0000 (1 + 0.08)^{60}$
c. $15,0000 [1/(1+0.02)^{60}]$ ✓
d. $15,0000 + (15000*0.05*30)$
17. Returns the interest payment for an investment for a given period?
- a. IPMT ✓
b. NPER
c. ISPMT
d. None of these.
18. In EXCEL function: ROUND(number,num_digits), which of the following could not be first argument?
- a. A range reference ✓
b. A number
c. a mathematical expression
d. A cell reference
19. You put \$1000 into an investment yielding 6% annual interest; you left the money in for two years. How much interest do you get at the end of those two years?
- a. $(1000)/(0.06)*2$
b. $(1000)/(12)$
c. $(1000)(0.06)(2)$ ✓
d. $(1000)(6)(2)$
20. On dividing the binomial; $x + 1$ by a monomial; x we get
- a. $(1/x) + 1$ ✓
b. $1/2$
c. 2
d. $1/x$

1. Range of data: 4,3,2,1,0,-1,-2,-3,-4 is

- a. 4
- b. -8
- c. 8 VU-HB
- d. -4

2. The sample variance is divided by.

- a. N-1
- b. n-1 VU-HB
- c. n
- d. N

3. In regression analysis, when we plot the values of dependent and independent variables, the resulting set of points is called

- a. Venn diagram
- b. Pie Graph
- c. Histogram
- d. Scatter Diagram VU-HB

4. A data in which we study about Regions is called

- a. Quantitative @hamzabahti25
- b. Qualitative
- c. Chronological
- d. Geographical VU-HB

5. Measures of dispersion indicate the

- a. relationships between variables.
- b. average score.
- c. central tendency of scores.
- d. extent to which scores differ from one another. VU-HB

6. Positive square root of the squared deviations of the values of a variance from their mean value is called.....

- a. Semi-inter quartile range
- b. Average deviation
- c. Variance

d. Standard deviation VU-HB

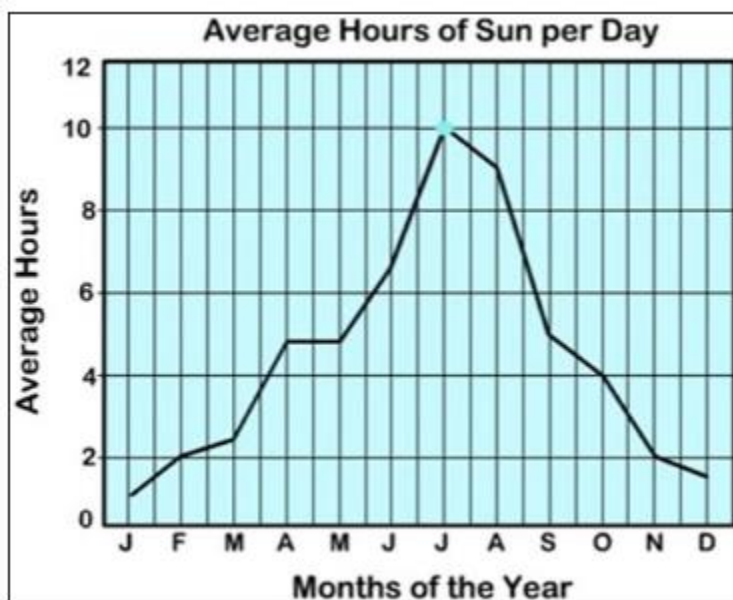
7. Ogives are used to determine at different values.

- a. Frequency distribution
- b. None of these

c. The cumulative Frequency VU-HB

- d. Frequency

8. The graph below shows the average hours of sun per day over a period of one year. Which other month had the same number of average hours of sun per day as February?



- a. November
- b. January
- c. July VU-HB**

d. December

9. The heights in centimeters of 5 students are:

165, 175, 176, 159, 170.

The sample median and sample mean are respectively:

a. 169, 170

b. 170, 169 VU-HB

c. 170, 170

d. 176, 169

10. An observation is measured in kilograms than the unit of standard deviation is.

a. Gram

b. Kilogram VU-HB

c. None of these

d. Kilogram²

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